

From Potential Difference to $\mathbf{E} = -\nabla\phi$

1) Start only from the definition of potential difference

For electrostatics, the potential difference between two points is

$$\phi(\mathbf{r}_2) - \phi(\mathbf{r}_1) = - \int_{\mathbf{r}_1}^{\mathbf{r}_2} \mathbf{E}(\mathbf{r}) \cdot d\mathbf{l} \quad (\text{path independent}).$$

We will now choose $\mathbf{r}_2 = \mathbf{r} + d\mathbf{l}$ extremely close to $\mathbf{r}$ and extract a local relation between $\mathbf{E}$ and the rate of change of ϕ .

2) A clean local geometric picture

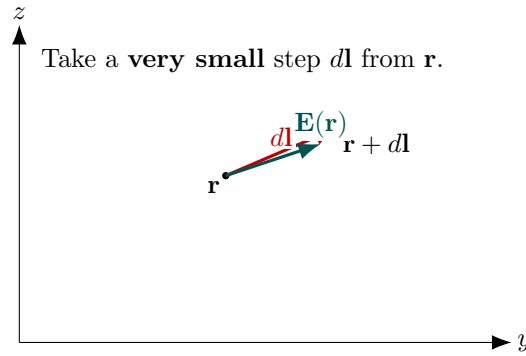


Figure 1: Local step from $\mathbf{r}$ to $\mathbf{r} + d\mathbf{l}$. For an infinitesimal displacement, the field is approximately constant on that tiny segment.

3) Evaluate the potential change over an infinitesimal step

Apply the potential-difference definition from $\mathbf{r}$ to $\mathbf{r} + d\mathbf{l}$:

$$\phi(\mathbf{r} + d\mathbf{l}) - \phi(\mathbf{r}) = - \int_{\mathbf{r}}^{\mathbf{r} + d\mathbf{l}} \mathbf{E}(\mathbf{r}') \cdot d\mathbf{l}'.$$

Because the segment is infinitesimally small, $\mathbf{E}(\mathbf{r}') \approx \mathbf{E}(\mathbf{r})$ on that segment, so

$$\phi(\mathbf{r} + d\mathbf{l}) - \phi(\mathbf{r}) \approx -\mathbf{E}(\mathbf{r}) \cdot \int_{\mathbf{r}}^{\mathbf{r} + d\mathbf{l}} d\mathbf{l}' = -\mathbf{E}(\mathbf{r}) \cdot d\mathbf{l}.$$

Thus,

$$d\phi \equiv \phi(\mathbf{r} + d\mathbf{l}) - \phi(\mathbf{r}) \approx -\mathbf{E}(\mathbf{r}) \cdot d\mathbf{l} \quad (\text{first-order change}).$$

4) Directional derivative viewpoint (extracting “rate of change along a direction”)

Let the small step be taken along a *unit* direction $\hat{\mathbf{a}}$:

$$d\mathbf{l} = \hat{\mathbf{a}} d\ell, \quad d\ell = \|d\mathbf{l}\|.$$

Then the previous relation becomes

$$d\phi \approx -\mathbf{E}(\mathbf{r}) \cdot \hat{\mathbf{a}} d\ell.$$

Divide by $d\ell$ and take the limit $d\ell \rightarrow 0$:

$$\lim_{d\ell \rightarrow 0} \frac{\phi(\mathbf{r} + \hat{\mathbf{a}} d\ell) - \phi(\mathbf{r})}{d\ell} = -\mathbf{E}(\mathbf{r}) \cdot \hat{\mathbf{a}}.$$

The left-hand side is precisely the **directional derivative** of ϕ along $\hat{\mathbf{a}}$, denoted

$$\left. \frac{\partial \phi}{\partial \ell} \right|_{\hat{\mathbf{a}}} \quad \text{or} \quad D_{\hat{\mathbf{a}}} \phi(\mathbf{r}).$$

Hence,

$$D_{\hat{\mathbf{a}}} \phi(\mathbf{r}) = -\mathbf{E}(\mathbf{r}) \cdot \hat{\mathbf{a}}.$$

5) Now identify the unique vector that reproduces all directional derivatives

A scalar field $\phi(\mathbf{r})$ has, at each point, a special vector $\nabla \phi(\mathbf{r})$ such that

$$D_{\hat{\mathbf{a}}} \phi(\mathbf{r}) = \nabla \phi(\mathbf{r}) \cdot \hat{\mathbf{a}} \quad \text{for every unit direction } \hat{\mathbf{a}}.$$

Comparing with the electrostatic result above,

$$\nabla \phi(\mathbf{r}) \cdot \hat{\mathbf{a}} = -\mathbf{E}(\mathbf{r}) \cdot \hat{\mathbf{a}} \quad \text{for every } \hat{\mathbf{a}}.$$

The only way two vectors have equal dot product with *every* direction $\hat{\mathbf{a}}$ is that the vectors are equal. Therefore,

$$\mathbf{E}(\mathbf{r}) = -\nabla \phi(\mathbf{r}).$$

6) Equipotential picture (why gradient direction is natural)

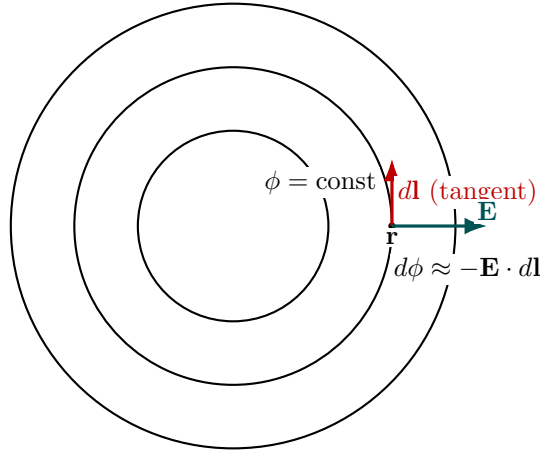


Figure 2: If $d\mathbf{l}$ is tangent to an equipotential curve, then $d\phi = 0$ and hence $\mathbf{E} \cdot d\mathbf{l} = 0$. So $\mathbf{E}$ is normal to equipotentials (same direction as $-\nabla\phi$).

Key Takeaways

- From the definition of potential difference, for an infinitesimal step:

$$d\phi \approx -\mathbf{E}(\mathbf{r}) \cdot d\mathbf{l}.$$

- Along a unit direction $\hat{\mathbf{a}}$, the directional derivative satisfies:

$$D_{\hat{\mathbf{a}}}\phi(\mathbf{r}) = -\mathbf{E}(\mathbf{r}) \cdot \hat{\mathbf{a}}.$$

- The unique vector reproducing all directional derivatives is the gradient, hence:

$$\boxed{\mathbf{E} = -\nabla\phi.}$$

- $\mathbf{E}$ is perpendicular to equipotential curves/surfaces.

Alternative (Geometric) Derivation of $\mathbf{E} = -\nabla\phi$

1) Two nearby equipotentials and a small displacement

Consider two nearby equipotential surfaces:

$$\phi = \phi_0 \quad \text{and} \quad \phi = \phi_0 + d\phi.$$



At a point, let $\hat{\mathbf{n}}$ be the unit normal to the equipotential surface (pointing toward increasing ϕ). Take an arbitrary small displacement $d\mathbf{l}$ that makes angle θ with $\hat{\mathbf{n}}$.

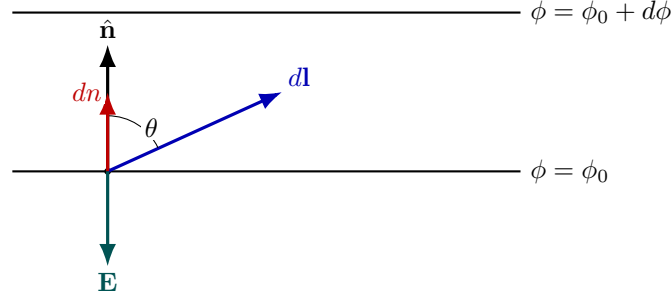


Figure 3: Equipotentials and a small displacement. The normal component is $dn = d\ell \cos \theta$. The electric field points toward decreasing potential.

2) Use the potential difference definition on an infinitesimal step

For an infinitesimal displacement,

$$d\phi \approx -\mathbf{E} \cdot d\mathbf{l}.$$

Since only the component of $d\mathbf{l}$ along $\mathbf{E}$ contributes to the dot product, and $\mathbf{E}$ is normal to equipotentials, write

$$\mathbf{E} = -E \hat{\mathbf{n}} \quad (E = \|\mathbf{E}\| \geq 0).$$

Then

$$d\phi \approx -(-E \hat{\mathbf{n}}) \cdot d\mathbf{l} = E (\hat{\mathbf{n}} \cdot d\mathbf{l}).$$

But the normal projection is

$$\hat{\mathbf{n}} \cdot d\mathbf{l} = d\ell \cos \theta \equiv dn.$$

Hence,

$$d\phi = E dn.$$

3) Directional derivative along the normal

Divide by dn and take the limit $dn \rightarrow 0$:

$$\frac{d\phi}{dn} = E.$$

Because $\mathbf{E}$ points opposite to increasing ϕ , the vector form is

$$\mathbf{E} = -\frac{d\phi}{dn} \hat{\mathbf{n}}.$$

Key result: the field magnitude equals the rate at which potential increases per unit distance in the normal direction.



4) Convert the “normal rate” into the full differential

Now write the total differential of $\phi(x, y, z)$:

$$d\phi = \frac{\partial\phi}{\partial x} dx + \frac{\partial\phi}{\partial y} dy + \frac{\partial\phi}{\partial z} dz.$$

Also, from $d\phi = -\mathbf{E} \cdot d\mathbf{l}$ with $d\mathbf{l} = dx \hat{\mathbf{x}} + dy \hat{\mathbf{y}} + dz \hat{\mathbf{z}}$,

$$d\phi = -(E_x dx + E_y dy + E_z dz).$$

Comparing coefficients of dx, dy, dz (true for arbitrary small displacements), we get

$$E_x = -\frac{\partial\phi}{\partial x}, \quad E_y = -\frac{\partial\phi}{\partial y}, \quad E_z = -\frac{\partial\phi}{\partial z}.$$

Collecting them into one vector equation,

$$\mathbf{E} = -\nabla\phi, \quad \nabla\phi = \hat{\mathbf{x}} \frac{\partial\phi}{\partial x} + \hat{\mathbf{y}} \frac{\partial\phi}{\partial y} + \hat{\mathbf{z}} \frac{\partial\phi}{\partial z}.$$

Key Takeaways

- $d\phi = -\mathbf{E} \cdot d\mathbf{l}$ (infinitesimal form of potential difference).
- Along the equipotential normal: $dn = d\ell \cos \theta$ and $d\phi = E dn$.
- Hence $\mathbf{E} = -(d\phi/dn) \hat{\mathbf{n}}$: field points toward decreasing potential.
- Expanding $d\phi$ in (x, y, z) gives $E_x = -\partial\phi/\partial x$, etc., i.e. $\mathbf{E} = -\nabla\phi$.

Deriving $\mathbf{E} = -\nabla\Phi$ in Spherical Coordinates

1) Coordinate transformation (Cartesian $\leftrightarrow$ spherical)

Use spherical coordinates (r, θ, φ) with

$$0 \leq \theta \leq \pi \quad (\text{polar angle from } +z), \quad 0 \leq \varphi < 2\pi \quad (\text{azimuth in the } xy\text{-plane}).$$

The Cartesian coordinates are

$$x = r \sin \theta \cos \varphi, \quad y = r \sin \theta \sin \varphi, \quad z = r \cos \theta.$$

2) “Translation” to dx, dy, dz (differentials in spherical)

Differentiate the relations above:

$$\begin{aligned} dx &= \sin \theta \cos \varphi dr - r \cos \theta \cos \varphi d\theta - r \sin \theta \sin \varphi d\varphi, \\ dy &= \sin \theta \sin \varphi dr + r \cos \theta \sin \varphi d\theta + r \sin \theta \cos \varphi d\varphi, \\ dz &= \cos \theta dr - r \sin \theta d\theta. \end{aligned}$$



3) Infinitesimal displacement vector in spherical form

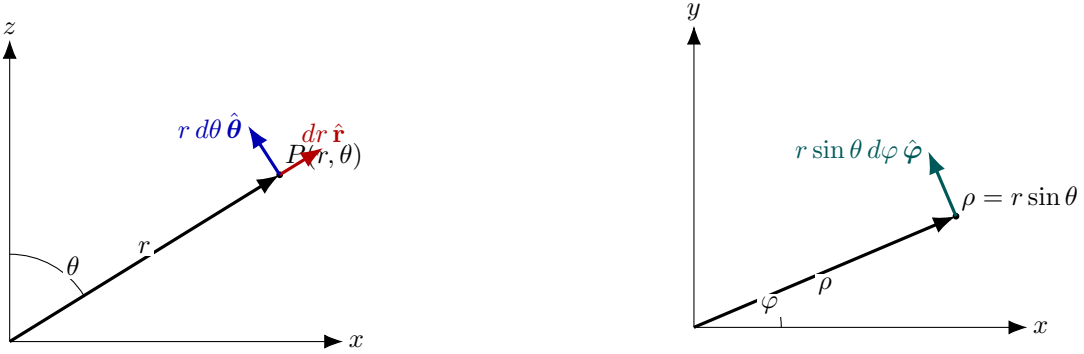
The infinitesimal displacement is

$$d\mathbf{l} = dx \hat{\mathbf{x}} + dy \hat{\mathbf{y}} + dz \hat{\mathbf{z}}.$$

In spherical coordinates, the same displacement decomposes along the local orthonormal directions $\{\hat{\mathbf{r}}, \hat{\boldsymbol{\theta}}, \hat{\boldsymbol{\varphi}}\}$ as

$$d\mathbf{l} = \hat{\mathbf{r}} dr + \hat{\boldsymbol{\theta}} (r d\theta) + \hat{\boldsymbol{\varphi}} (r \sin \theta d\varphi).$$

(So the spherical scale factors are $h_r = 1$, $h_\theta = r$, $h_\varphi = r \sin \theta$.)



In the xz -plane: dr along $\hat{\mathbf{r}}$, and $r d\theta$ along $\hat{\boldsymbol{\theta}}$.

In the xy -plane: azimuthal step has length $r \sin \theta d\varphi$.

Figure 4: Spherical differential displacement: $d\mathbf{l} = \hat{\mathbf{r}} dr + \hat{\boldsymbol{\theta}} (r d\theta) + \hat{\boldsymbol{\varphi}} (r \sin \theta d\varphi)$.

4) Gradient derivation in spherical coordinates

Let the electrostatic potential be $\Phi(r, \theta, \varphi)$ (using capital Φ to avoid confusion with the azimuth φ). Its total differential is

$$d\Phi = \frac{\partial \Phi}{\partial r} dr + \frac{\partial \Phi}{\partial \theta} d\theta + \frac{\partial \Phi}{\partial \varphi} d\varphi.$$

From the infinitesimal potential relation,

$$d\Phi = -\mathbf{E} \cdot d\mathbf{l}.$$

Write the field in spherical components:

$$\mathbf{E} = E_r \hat{\mathbf{r}} + E_\theta \hat{\boldsymbol{\theta}} + E_\varphi \hat{\boldsymbol{\varphi}},$$

and substitute $d\mathbf{l}$ from Section 3:

$$\mathbf{E} \cdot d\mathbf{l} = (E_r \hat{\mathbf{r}} + E_\theta \hat{\boldsymbol{\theta}} + E_\varphi \hat{\boldsymbol{\varphi}}) \cdot (\hat{\mathbf{r}} dr + \hat{\boldsymbol{\theta}} (r d\theta) + \hat{\boldsymbol{\varphi}} (r \sin \theta d\varphi)).$$

Using orthonormality $\hat{\mathbf{r}} \cdot \hat{\mathbf{r}} = 1$, $\hat{\boldsymbol{\theta}} \cdot \hat{\boldsymbol{\theta}} = 1$, $\hat{\boldsymbol{\varphi}} \cdot \hat{\boldsymbol{\varphi}} = 1$, and cross-terms = 0, we get

$$\mathbf{E} \cdot d\mathbf{l} = E_r dr + E_\theta r d\theta + E_\varphi r \sin \theta d\varphi.$$



Therefore,

$$d\Phi = -(E_r dr + E_\theta r d\theta + E_\varphi r \sin \theta d\varphi).$$

Now compare coefficients of the independent differentials dr , $d\theta$, $d\varphi$:

$$E_r = -\frac{\partial \Phi}{\partial r}, \quad E_\theta = -\frac{1}{r} \frac{\partial \Phi}{\partial \theta}, \quad E_\varphi = -\frac{1}{r \sin \theta} \frac{\partial \Phi}{\partial \varphi}.$$

5) Gradient in spherical form (result)

Collecting these into one vector expression,

$$\nabla \Phi = \hat{\mathbf{r}} \frac{\partial \Phi}{\partial r} + \hat{\boldsymbol{\theta}} \frac{1}{r} \frac{\partial \Phi}{\partial \theta} + \hat{\boldsymbol{\varphi}} \frac{1}{r \sin \theta} \frac{\partial \Phi}{\partial \varphi}, \quad \mathbf{E} = -\nabla \Phi.$$

Example: Electric Field of a Dipole (at the origin) from the Potential (Spherical Coordinates)

Setup and symmetry

Place $+q$ at $z = +a$ and $-q$ at $z = -a$ on the z -axis. The dipole moment is

$$\mathbf{p} = q(2a) \hat{\mathbf{z}}, \quad p = 2aq.$$

Choose an observation point $P(r, \theta, \varphi)$ in spherical coordinates, where θ is the polar angle from $+z$.

Why there is no φ dependence (intuitive symmetry) $\Rightarrow E_\varphi = 0$

Rotate the entire configuration about the z -axis by any angle $\Delta\varphi$:

- The charges at $(0, 0, \pm a)$ remain unchanged.
- The distance from P to each charge depends only on r and the polar angle θ , not on the azimuth φ .

Therefore the potential must satisfy

$$\Phi(r, \theta, \varphi) = \Phi(r, \theta),$$

so

$$\frac{\partial \Phi}{\partial \varphi} = 0 \quad \Rightarrow \quad E_\varphi = -\frac{1}{r \sin \theta} \frac{\partial \Phi}{\partial \varphi} = 0.$$

Physical intuition: the system has perfect axial (cylindrical) symmetry about z , so there is no preferred “twisting” direction around the axis; hence no $\hat{\boldsymbol{\varphi}}$ component.

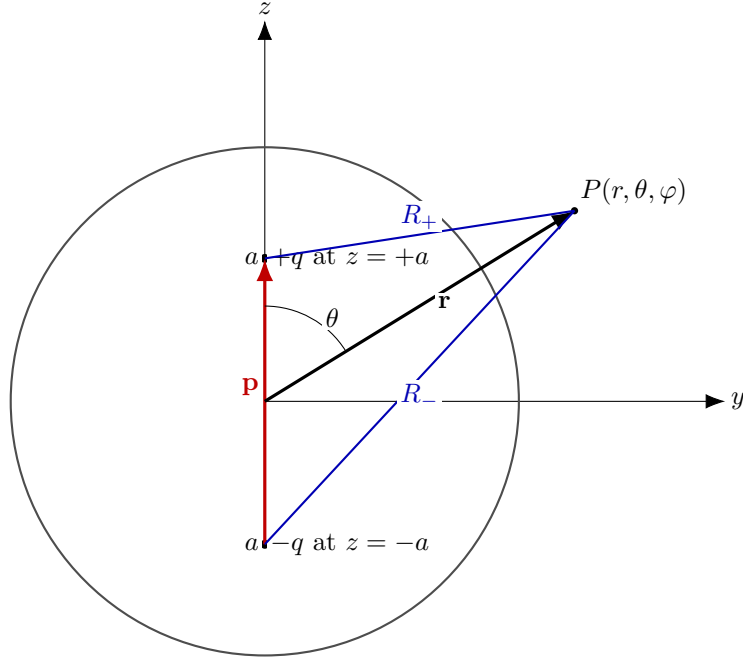


Figure 5: Dipole on the z -axis with observation point $P(r, \theta, \varphi)$. Distances to charges are R_+ and R_- .

Potential from geometry (exact for the two charges)

By superposition,

$$\Phi(P) = \frac{1}{4\pi\epsilon_0} \left(\frac{q}{R_+} - \frac{q}{R_-} \right).$$

Now compute $R_{\pm}$ purely from geometry. Using the law of cosines with the triangle formed by $\mathbf{r}$ and $\pm a\hat{\mathbf{z}}$:

$$R_{\pm}^2 = r^2 + a^2 \mp 2ar \cos \theta \quad \Rightarrow \quad R_{\pm} = \sqrt{r^2 + a^2 \mp 2ar \cos \theta}.$$

Hence the **exact** potential is

$$\Phi(r, \theta) = \frac{q}{4\pi\epsilon_0} \left(\frac{1}{\sqrt{r^2 + a^2 - 2ar \cos \theta}} - \frac{1}{\sqrt{r^2 + a^2 + 2ar \cos \theta}} \right).$$

Dipole (far-field) form from the same geometry

When $r \gg a$, the point is far compared to separation. The potential simplifies to the standard dipole form:

$$\Phi(r, \theta) \approx \frac{1}{4\pi\epsilon_0} \frac{p \cos \theta}{r^2}, \quad (r \gg a, \quad p = 2aq).$$

(You may keep the exact expression above for general r , but the closed-form field is simplest in the dipole limit.)



Electric field from the spherical gradient

Using spherical-coordinate relations (from the previous section):

$$E_r = -\frac{\partial \Phi}{\partial r}, \quad E_\theta = -\frac{1}{r} \frac{\partial \Phi}{\partial \theta}, \quad E_\varphi = 0.$$

Using the dipole form $\Phi = \frac{1}{4\pi\epsilon_0} \frac{p \cos \theta}{r^2}$:

$$\frac{\partial \Phi}{\partial r} = \frac{1}{4\pi\epsilon_0} p \cos \theta \frac{\partial}{\partial r} (r^{-2}) = \frac{1}{4\pi\epsilon_0} p \cos \theta (-2)r^{-3} = -\frac{1}{4\pi\epsilon_0} \frac{2p \cos \theta}{r^3},$$

so

$$E_r = -\frac{\partial \Phi}{\partial r} = \frac{1}{4\pi\epsilon_0} \frac{2p \cos \theta}{r^3}.$$

Next,

$$\frac{\partial \Phi}{\partial \theta} = \frac{1}{4\pi\epsilon_0} \frac{p}{r^2} \frac{\partial}{\partial \theta} (\cos \theta) = -\frac{1}{4\pi\epsilon_0} \frac{p \sin \theta}{r^2},$$

thus

$$E_\theta = -\frac{1}{r} \frac{\partial \Phi}{\partial \theta} = \frac{1}{4\pi\epsilon_0} \frac{p \sin \theta}{r^3}.$$

Final dipole field (spherical components)

$$\mathbf{E}(r, \theta) = E_r \hat{\mathbf{r}} + E_\theta \hat{\boldsymbol{\theta}} = \frac{1}{4\pi\epsilon_0} \frac{1}{r^3} \left(2p \cos \theta \hat{\mathbf{r}} + p \sin \theta \hat{\boldsymbol{\theta}} \right), \quad E_\varphi = 0.$$

Key Takeaways

- Axial symmetry about z implies Φ does not depend on $\varphi \Rightarrow E_\varphi = 0$.
- Geometry gives $R_\pm = \sqrt{r^2 + a^2 \mp 2ar \cos \theta}$ and the exact potential

$$\Phi = \frac{q}{4\pi\epsilon_0} \left(\frac{1}{R_+} - \frac{1}{R_-} \right).$$

- Far from the dipole ($r \gg a$): $\Phi \approx \frac{1}{4\pi\epsilon_0} \frac{p \cos \theta}{r^2}$.
- Using spherical gradient: $\mathbf{E} = \frac{1}{4\pi\epsilon_0 r^3} \left(2p \cos \theta \hat{\mathbf{r}} + p \sin \theta \hat{\boldsymbol{\theta}} \right)$.